

BRIAN QIAN

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github.com/Azlothe

Education

University of Texas at Austin

Master of Science, Computer Science

Expected June 2027

GPA: 4.00/4.00

San Jose State University

Bachelor of Science, Computer Science

Graduated June 2024

GPA: 4.00/4.00

Experience

Company A

Start Date — Present

Software Engineer

Start Date – Present

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Software Engineer Intern

Start Date – End Date

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Company B

Start Date — End Date

Software Engineer Intern

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Projects

LUCM (Large Unit Conversion Model) | Python, PyTorch, LoRA, RLHF, Transformers

March 2026

- Fine-tuned a 1.7B parameter SmolLM2 model with **LoRA adapters**, further augmenting it with in-context learning and Chain-of-Thought reasoning to generate high-quality reasoning traces for unit conversion tasks.
- Distilled the 1.7B model into a compact 360M parameter SmolLM2 model using **Rejection Sampling Fine-Tuning (RFT)**, achieving 83% reasoning accuracy while reducing model size by approximately 80%.

GPU-Accelerated Clustering | C++, GPU, CUDA, Thrust, Nsight Compute

August 2025 — October 2025

- Developed CPU & GPU implementations of the K-Means clustering algorithm, achieving an 80x speedup on GPU.
- Profiled GPU performance and identified performance bottlenecks of data transfer cost and frequency, synchronization, expensive memory accesses & cache misses, thread granularity, queued kernel launches, and warp scheduling.
- Optimized **CUDA** performance using shared memory, streams, atomics, tiling, and coalesced memory accesses.

Astrophysical N-Body Simulation | C++, MPI, Distributed Systems, High-Performance Computing

December 2025

- Built a distributed N-body simulator with **MPI**, partitioning 100K+ bodies across distributed nodes with replicated quadrees and ring-based state exchange to eliminate centralized computation bottlenecks and serialization overhead.
- Reduced memory footprint by 80% (1.5 GB → 287 MB) after memory profiling with **valgrind**, identifying optimizations for the dynamic quadrees through lazy quadrant allocation, incremental mass propagation, and precomputation.
- Investigated the communication-vs-computation tradeoff inherent in my design, analyzing empirical parallel performance to identify the inflection point where synchronization latency dominates computation throughput.

Hobbies & Interests: ...